MICROBES IN HUMAN WELFARE

The A+ Scholar's Biological Master-Guide

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Expert Note: Microbes aren't just pathogens! This notebook maps how we harness bacteria, fungi, and viruses for human survival and industry.

Household Products: The Chemistry of Cheese

Swiss Cheese

- Feature: Known for large holes.
- Mechanism: Holes are due to the production of a large amount of CO_2.
- Microbe Responsible: Propionibacterium sharmanii (Bacterium).

Roquefort Cheese

- Feature: Known for a specific, distinct flavour.
- Mechanism: Ripened by growing a specific fungi on them.

Expert Note: In cheese making, the specific microbe dictates the texture (gas pockets) and the taste profile!

Microbes in Household Products: Dairy & Dough

Dairy – The LAB Process

Lactic Acid Bacteria (LAB) converting milk to curd.

Lactobacillus

Inoculum/starter added to milk
↓
LAB multiplies at suitable temperature
↓
Produces acids
↓
Acids coagulate & partially digest milk proteins.

Benefits:

1. Improves nutritional quality by increasing Vitamin B12.
2. Checks disease-causing microbes in our stomach.

Dough & Traditional Foods

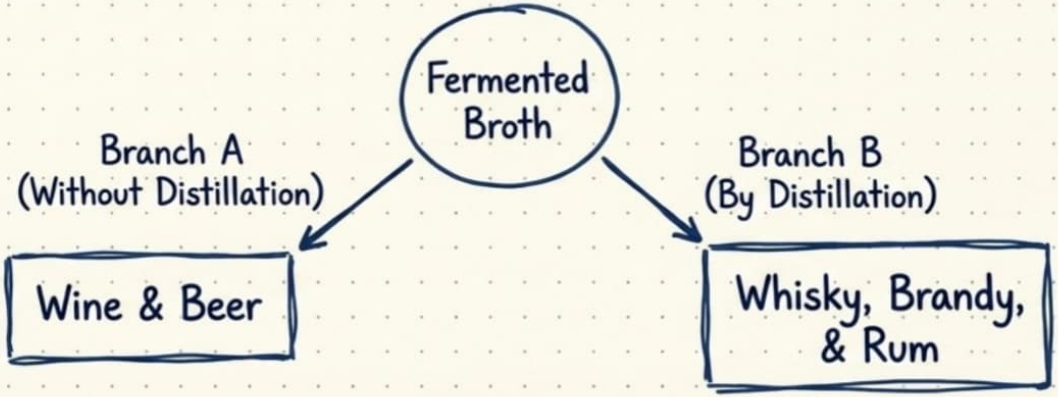
- Idli & Dosa: Dough fermented by bacteria. CO_2 production makes it puffy.
- Bread: Dough fermented by baker's yeast (Saccharomyces cerevisiae).
- Toddy: Traditional South Indian drink made by fermenting sap from palm trees.
- Other Foods: Microbes also used to ferment fish, soybean & bamboo shoots.

Industrial Products: Fermented Beverages

Context: Production on an industrial scale requires growing microbes in very large vessels called Fermentors.

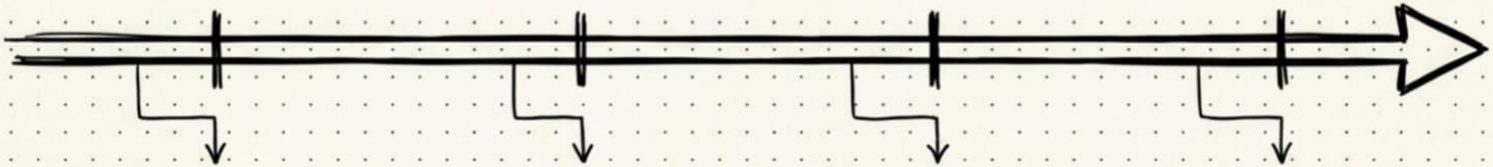
The Organism: Saccharomyces cerevisiae (also called brewer's yeast).

Input: Used for fermenting malted cereals & fruit juices to produce ethanol.



Health & Medicine: Antibiotics & The Penicillin Timeline

Etymology:
"Anti" (Greek: against) + "bio" (life) = "Against Life" (in the context of disease-causing organisms).



The Discovery (Alexander Fleming): Working on *Staphylococci* bacteria. Observed a mould growing in an unwashed culture plate. Chemical produced was named Penicillin after the mould *Penicillium notatum*.

The Establishment (Ernest Chain & Howard Florey): Established its full potential as an effective antibiotic.

World War II: Extensively used to treat American soldiers during WWII.

The Recognition (1945): Fleming, Chain, and Florey awarded the Nobel Prize for this discovery.

Health & Medicine: The Scope of Antibiotics

Antibiotics are widely used for treating deadly diseases that once ravaged populations.

- Plague
- Whooping Cough (Kali Khansi)
- Diphtheria (Gal Ghotu)
- Leprosy (Kusht Rog)

Expert Note: The discovery of antibiotics transformed these formerly fatal diseases into highly treatable conditions.

Matrix 1: The Commercial Chemicals Ledger

Context: Microbes are used for commercial & industrial production of specific chemicals like organic acids and alcohols.

Microbe	Type	Product
Aspergillus niger	Fungus	Citric Acid
Acetobacter aceti	Bacteria	Acetic Acid
Clostridium butylicum	Bacteria	Butyric Acid
Lactobacillus	Bacteria	Lactic Acid
Saccharomyces cerevisiae	Yeast	Ethanol (Commercial production)

Matrix 2: Enzymes & Bioactive Molecules Profile

Product/Molecule	Source Microbe	Type	Application
Lipases	–	–	Used in detergent formulations; helpful in removing oily stains from laundry.
Pectinases & Proteases	–	–	Used to clarify bottled juices.
Streptokinase	Streptococcus	Bacterium	Used as a 'clot buster' for removing clots in blood vessels of myocardial infarction (heart attack) patients.
Cyclosporin A	Trichoderma polysporum	Fungus	Used as an immunosuppressive agent in organ-transplant patients.
Statins	Monascus purpureus	Yeast	Commercial blood-cholesterol lowering agent (inhibits enzymes responsible for cholesterol synthesis).

Environmental Welfare: Sewage Treatment in STPs (Stage 1)

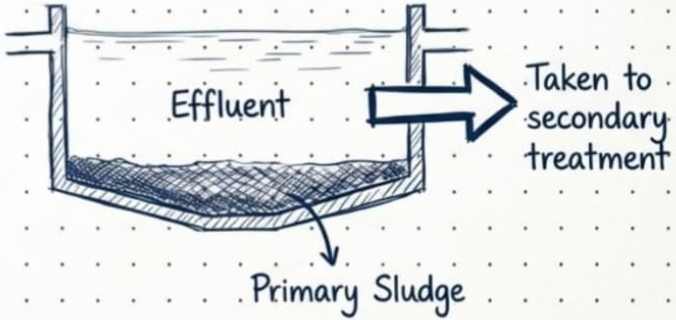
Waste water generated everyday contains a large amount of human excreta. This is called Sewage. It contains large amounts of organic matter & microbes.

Treated in Sewage Treatment Plants (STPs) by heterotrophic microbes naturally present in the sewage.

Process Map (Primary Treatment - Physical Removal)

Step 1: Sequential Filtration ⟶ Removes floating debris.

Step 2: Sedimentation ⟶ Removes grit (soil & small pebbles).



The Complete STP Process Flow

Filtration / Sedimentation (Floating debris removed by sequential filtration) → Grit is removed by sedimentation → Air is pumped out & flocs produced → BOD reduced as flocs consume effluents → Flocs sediment (Activated Sludge) → Anaerobic bacteria digest already present bacteria & fungi in anaerobic sludge digester. Biogas produced.

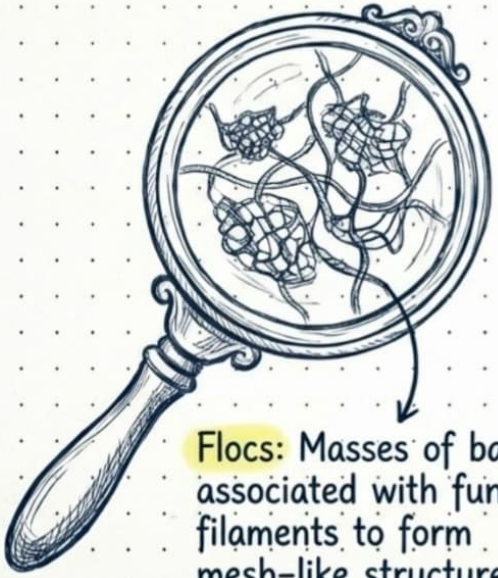
Supporting Details

- Activated Sludge: A small amount is pumped back to the aeration tank as inoculum. The remaining part pumped to large Anaerobic sludge digesters.
- Biogas: Anaerobic digestion produces gases (H_2S, CH_4, CO_2) called biogas. Effluent released into water bodies.

Action Plan: GAP & YAP
(Ministry of Environment & Forests initiated Ganga & Yamuna Action Plan to save rivers from pollution; large number of STPs will be built).

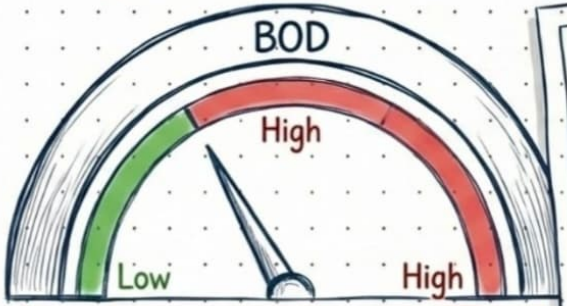
Environmental Welfare: Secondary (Biological) Treatment & BOD

The Aeration Process: Primary effluent is passed to large aeration tanks where air is pumped out. This leads to vigorous growth of microbes.



Flocs: Masses of bacteria associated with fungal filaments to form mesh-like structures.

While growing, flocs consume a major part of organic matter in the effluent.



Biochemical Oxygen Demand (BOD)

Refers to the amount of oxygen that would be consumed if all the organic matter in 1L of water were oxidized by bacteria.

Warning: Greater the BOD, more is its polluting potential.

Outcome: Sewage is treated till BOD is reduced.

Energy Production: Methanogens & Biogas

Core Principle: Type of gas produced depends upon the microbes & organic substrates utilized. (e.g., in dough/cheese, CO_2 is produced).

Focus: Methanogens

- Some bacteria grow anaerobically on cellulosic material to produce CO_2 & H_2. Collectively called methanogens.
- Scientific Name: Methanobacterium
- Habitat: Commonly found in anaerobic sludge of STPs and in the rumen (gut) of cattle.

Cattle Connection: Rumen bacteria help breakdown cellulose (important for cattle nutrition) → Excreta of cattle (dung) is highly rich in these bacteria → Dung is used to produce biogas.

Agricultural Welfare: Biocontrol Agents

The Organic Philosophy

Organic farmers work to create a system where pests are not eradicated but instead kept at manageable levels.

Warning: Eradication is undesirable! It kills useful insects and reduces food options for beneficial predatory insects that feed on them.

Simple Predators

- Ladybirds → Useful to get rid of aphids.

- Dragonflies → Useful to get rid of mosquitoes.

Bacterial Biocontrol

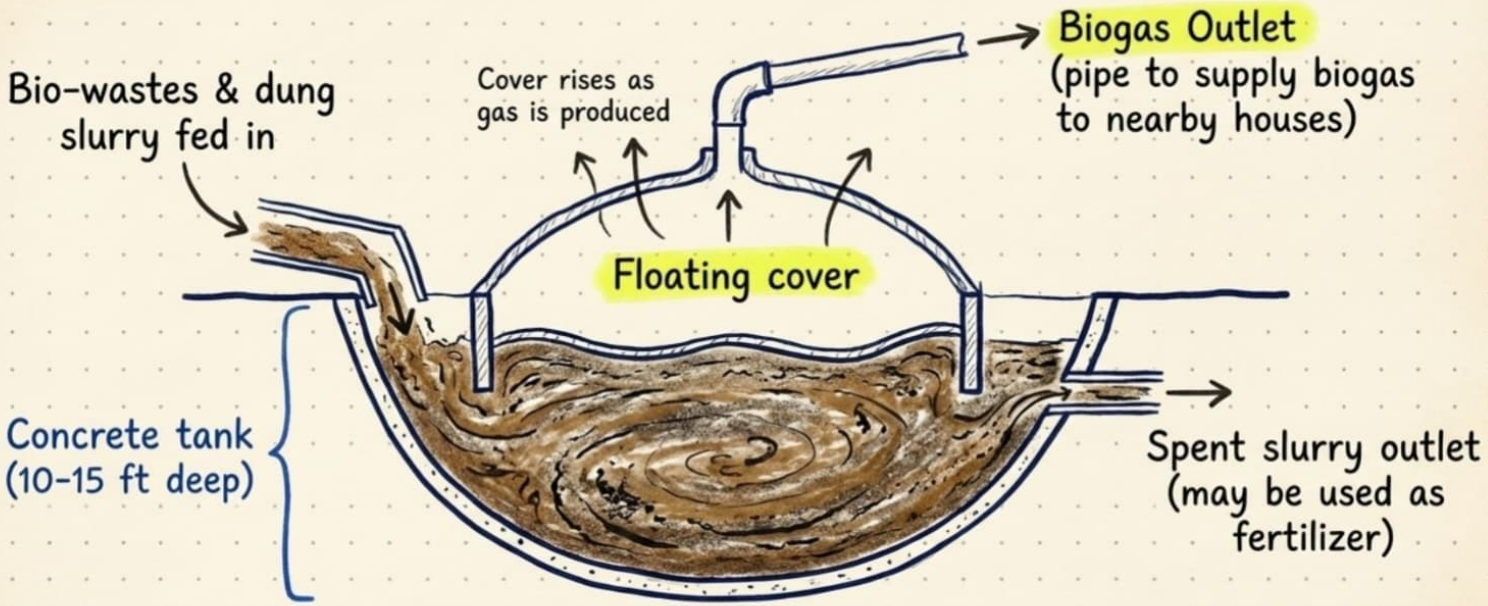
Bacillus thuringiensis (Bt)

Used to get rid of butterfly caterpillars on plants like brassicas & fruit trees.

Mechanism: Toxin released in the gut of the larvae kills them.

Application: Genes introduced into plants to make them insect-resistant (e.g., Bt cotton).

The Rural Biogas Plant Blueprint



Deployment: Built primarily in rural areas. Technology developed in India by:

1. Indian Agricultural Research Institute (IARI),
2. Khadi and Village Industries Commission (KVIC).

Agricultural Welfare: Biofertilisers (Bacteria & Fungi)

Organisms that enrich the nutrient quality of the soil. Main sources are bacteria, fungi,, and cyanobacteria.

Bacterial Nitrogen (N_2) Fixation

- Symbiotic: Nodules on roots of leguminous plants formed by symbiotic association of Rhizobium.
- Free-living: Other bacteria can also fix N_2 in the soil. Examples: Azospirillum & Azotobacter.

Fungal Symbiotic Association

- Term: Mycorrhiza (most common association formed by genus Glomus).
- Mechanism: The fungal symbiont absorbs phosphorus from the soil and passes it to the plant.
- Plant Benefits: Resistance to root-borne pathogens, tolerance to salinity & drought.

Matrix 3: Advanced Biocontrol Profiles

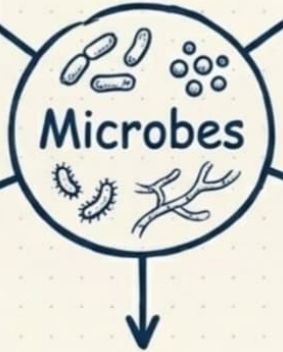
Agent 1: Root Fungi	Agent 2: Viruses
Microbe: *Trichoderma* Action: Helps in controlling pathogenic plant diseases. Location: Root ecosystems.	Microbe: Baculoviruses (Majorly in genus *Nucleopolyhedrovirus*) Target: Attack insects & other arthropods. Highly species-specific. They do not have any bad effects on plants, mammals, birds, fishes, or non-target insects.

Conclusion: Desirable when beneficial insects are being conserved to aid in an overall Integrated Pest Management (IPM) programme.

Synthesis: The Web of Microbial Welfare

Household: Nutrition & Fermentation (Lactobacillus, Yeast)

Industry & Health: Antibiotics & Enzymes (Penicillium, Statins)



Agriculture: Pest Control & Soil Fertility (Bt, Mycorrhiza)

Environment: Sewage Purification (Flocs, BOD Reduction)

Energy: Biogas Generation (Methanobacterium)

Expert Note: From digesting our milk to cleaning our cities, microbes are the invisible, indispensable engines of human welfare.

Biofertilisers: Cyanobacteria & Final Taxonomy

Cyanobacteria

- Autotrophic microbes capable of atmospheric N2 fixation.
- Examples: Anabaena, Nostoc, and Oscillatoria
- Application: Extensively used in paddy fields.
- Bonus: Blue-green algae also add organic matter to the soil and increase its fertility.

Matrix 4 - Biofertiliser Taxonomy

Category	Mechanism	Key Microbes
1 Bacteria (Symbiotic)	N2 Fixation in root nodules	Rhizobium
2 Bacteria (Free-living)	N2 Fixation in soil	Azospirillum, Azotobacter
3 Fungi (Symbiotic)	Phosphorus absorption (Mycorrhiza)	Glomus
4 Cyanobacteria	Atmospheric N2 Fixation / Organic matter	Anabaena, Nostoc, Oscillatoria